First- and Second-Order Optical Coherence as Probes of Spacetime Foam

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Quantum gravity is expected to render spacetime stochastic at the Planck scale. We show that the resulting random arrival-time jitter affects the field-correlation and intensity-correlation functions of light from distant sources differently, breaking the Gaussian-statistics identity (the Siegert relation) that links them for all chaotic-thermal light. For spatially extended sources, foam-induced jitter between rays from different parts of the source produces an additional suppression of the zero-lag photon-bunching (Hanbury Brown–Twiss) peak that depends on the source angular size and probes the transverse coherence length of the foam. Both signatures are achromatic and follow a distance scaling distinct from plasma scintillation or atmospheric turbulence, making them complementary to the existing constraints from imaging and from energy-dependent time-of-flight measurements. We develop the framework in the narrow-band, weak-jitter, point-source limit and its extension to spatially resolved sources, recover the published Strehl-ratio and fringe-visibility tests as limiting cases, and quantify the predicted observables for the Ng–Perlman family of foam models. For the holographic exponent ($\alpha = 2/3$), the framework sets upper limits on the line-of-sight foam path-length variance and on the transverse coherence length that approach but do not clearly exceed the systematic floors anticipated for next-generation optical intensity interferometers.

I. INTRODUCTION

A long-standing expectation from the union of quantum mechanics and general relativity is that the smooth pseudo-Riemannian manifold of classical spacetime breaks down at the Planck length $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616 \times 10^{-35}$ m. In Wheeler's original "spacetime foam" picture [1], quantum fluctuations of the metric on this scale render the geometry stochastic; similar conclusions are reached in the Euclidean path-integral treatment [2] and in modern holographic and entropic-gravity arguments from very different starting points. While a complete theory of quantum gravity remains elusive, the question of whether residual imprints of Planck-scale geometric fluctuations can accumulate over cosmological baselines and become observable is one of the few empirical handles on physics in the Planck regime, and it has motivated a substantial body of phenomenological work over the past two decades [3, 4].

A widely studied parametrization, due to Ng and collaborators [4–6], assigns the cumulative line-of-sight path-length variance acquired by light traversing a distance D the power-law form

$$\sigma_\ell^2(D) \;=\; A_\alpha\, D^{2(1-\alpha)}\, \ell_{\rm P}^{2\alpha}, \tag{1}$$

where A_α is a dimensionless coefficient of order unity and the exponent α encodes the underlying foam dynamics. Three values are conventionally distinguished. The "conservative" case $\alpha = 1$ follows from Wheeler's original dimensional argument and gives $\sigma_\ell \sim \ell_{\rm P}$ regardless of D – observationally negligible. The "holographic" case $\alpha = 2/3$ is motivated by the requirement that the information content of a region of spacetime saturate the Bekenstein–Hawking bound and scale with the area of its boundary in Planck units [5]; it predicts $\sigma_\ell \sim (D\ell_{\rm P}^2)^{1/3}$, of order 10^{-15} m over 1 Gpc. The "random-walk" case $\alpha = 1/2$ treats successive Planck cells as independent path-length kicks and gives $\sigma_\ell \sim (D\ell_{\rm P})^{1/2}$, roughly 10^{-5} m over the same distance – large enough to randomize the optical phase of light from sources at $z \gtrsim 1$. The three cases bracket the range investigated most thoroughly; for a recent review situating them within the broader quantum-gravity phenomenology program, see Carlip [3].

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The foam line of inquiry sits within a broader program of quantum-gravity phenomenology that has matured substantially in the multi-messenger era [7, 8]. Other channels covered in those reviews include searches for Lorentz-invariance violation in high-energy astrophysical transients, vacuum birefringence over cosmological baselines, and modifications of gravitational-wave propagation; complementary laboratory programs target Planck-scale metric noise via tabletop interferometers [9], and a parallel theoretical line asks whether quantum-gravitational degrees of freedom decohere otherwise classical photon states during propagation [10]. The present work belongs to the propagation-based, coherence-channel branch of this landscape: we ask what imprint the Ng–Perlman foam ansatz leaves on the first- and second-order temporal correlations of light from cosmologically distant sources.

Two complementary observational channels have been used to test these models. The first exploits the fact that random Planck-scale path-length deviations imprint a stochastic phase $\sigma_\phi = 2\pi\sigma_\ell/\lambda$ across the aperture of any telescope, reducing the Strehl ratio by $S \approx \exp(-\sigma_\phi^2)$ and degrading the diffraction-limited point-spread function. Detections of crisp, near-diffraction-limited images of distant active galactic nuclei with the Hubble Space Telescope and ground-based instruments [11–14], together with stellar interferometry, exclude $\alpha \lesssim 2/3$ at optical wavelengths for $D \sim 1$–10 Gpc, leaving the holographic and conservative cases as the only quantitatively viable models. These exclusions assume transverse foam correlations are negligible on aperture scales; models with long-range transverse correlations can evade blurring constraints while retaining large $\sigma_\ell(D)$ [15]. The second channel uses energy-dependent time-of-flight measurements of high-energy astrophysical transients — gamma-ray bursts observed by Fermi [16], TeV blazar flares from H.E.S.S. [17], and LHAASO observations of GRB 221009A [18], currently the leading limits — to constrain the Lorentz-invariance-violating (LIV) dispersion $\delta t \propto (E/E_{\rm QG})^n$ that some quantum-gravity models predict. A recent analysis of GRB 221009A claims consistency with holographic foam blurring [19], though without formal statistical significance. Together the two channels have winnowed the parameter space but not eliminated it, and several scalings — including the holographic $\alpha = 2/3$ — remain unconstrained.

Both channels share a structural limitation that motivates the present work. The imaging tests are essentially binary: once a near-diffraction-limited image is registered the constraint saturates, and no further information about the residual phase variance is recovered. The time-of-flight tests, conversely, target a deterministic, energy-ordered delay — precisely what is *not* expected from a geometric-optics path-length fluctuation $\delta t = \Delta\ell/c$, which is frequency-independent. The stochastic, achromatic spacetime foam jitter considered here therefore leaves no LIV signal, and a foam model whose phase variance lies below the imaging threshold is invisible to both channels. We address the natural question of whether observables exist that bridge this gap — sensitive to a phase variance too small to erase fringes, but achromatic and thus distinguishable from plasma scintillation or instrumental dispersion.

We develop such a framework based on the first-order (field–field) and second-order (intensity–intensity) coherence functions of light from astrophysical sources, denoted $g^{(1)}(\tau)$ and $g^{(2)}(\tau)$ in standard quantum-optics notation. The first-order coherence is equivalently the fringe visibility measured in a stellar interferometer at delay τ; the second-order coherence is the intensity correlation measured in a Hanbury Brown–Twiss experiment [20] (intensity correlations between two photon-counting detectors, originally developed for stellar angular-diameter measurements). For all stationary chaotic-thermal sources the two are linked by the Siegert relation [21], the Gaussian-statistics identity $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$. The framework recovers the published imaging and fringe-visibility tests as limiting cases of a single first-order observable, and we identify two additional foam signatures inaccessible to those tests: one accessible to single-aperture temporal coherence measurements on point-like sources, and one to intensity interferometry on spatially resolved sources such as that recently demonstrated by VERITAS [22, 23]. Their derivation, observability, and use as discriminants against conventional propagation effects are the subject of this paper.

The physical picture is simple: in the narrow-band limit, spacetime foam jitter shifts the arrival time of a photon by δt without appreciably changing its intensity envelope. The first-order coherence $g^{(1)}$ measures field–field correlations and so picks up the carrier-phase modulation $e^{i\omega_0\delta t}$, which is exponentially suppressed when the rms phase $\sigma_\phi = \omega_0\langle\delta t^2\rangle^{1/2}$ exceeds unity. The second-order coherence $g^{(2)}$ measures intensity–intensity correlations and is blind to that carrier phase. The two observables therefore decouple, and the algebraic relation between them — the Siegert relation — is broken in a foam-specific way.

The second-order observable proposed here is already routinely computed in radio astronomy under different names (spectral kurtosis, modulation index; Section VIII A), so the framework translates directly to existing data products.

The remainder of the paper is organized as follows. Section II develops the model: the spatiotemporal covariance of the foam-induced path-length fluctuation field, the differential-jitter distribution at a single ray, and its generalization to ray pairs. Section III treats the first-order coherence and its connection to the published Strehl-ratio and fringe-visibility constraints. Section IV treats the second-order coherence, including its conditional and unconditional forms in the narrow-band limit. Section V examines the Siegert relation under the foam-modified field statistics. Section VI extends the analysis to spatially resolved sources, derives the suppression of the zero-lag HBT bunching peak, and gives an inversion procedure for the transverse foam coherence length $r_\perp$. Section VII identifies the distance scaling and achromaticity that distinguish foam from conventional propagation phase screens, and discusses the self-calibrating character of the second-order observables — features shared by both the point-source and extended-source signatures.

Section VIII concludes.

II. MODEL

A. Arrival-Time Jitter as a Time Reparameterization

We model the detected field as a stochastically time-warped version of the intrinsic source field:

$$E_{\rm obs}(t) = E_{\rm src}(t - \delta t(t))\,, \tag{2}$$

where $E_{\rm src}(t)$ is the complex amplitude of the light at the source — modelled as a stationary zero-mean complex Gaussian process, the standard description of a single chaotic/thermal mode — and $\delta t(t)$ is the small, time-dependent shift in arrival time picked up by a photon as it traverses fluctuating Planck-scale geometry, treated as a stationary stochastic process statistically independent of $E_{\rm src}$. The jitter arises from cumulative path-length fluctuations along the line of sight of propagation distance D:

$$\delta t(t) = \frac{1}{c}\,\Delta\ell_{\rm tot}(t). \tag{3}$$

In the geometric-optics limit the path-length shift $\Delta\ell_{\rm tot}$ is a purely geometric quantity and hence frequency-independent (achromatic), in contrast to plasma dispersion which scales as λ^2. This achromaticity is a key discriminant between foam and conventional propagation effects.

B. Spatial and Temporal Fluctuations of Spacetime Foam

Because foam is a property of spacetime rather than of any preferred direction, the same dimensional logic that governs longitudinal path-length fluctuations applies equally to transverse baselines. By dimensional analogy with the longitudinal scaling, the transverse coherence has a characteristic length scale $r_\perp$. We do not attempt to fix its value from first principles — the longitudinal scaling counts cells traversed along a ray, whereas the transverse case mixes both the in-line distance D and the cross-ray separation b in a way that is qualitatively distinct from the single-ray accumulation — and instead treat $r_\perp$ as a free parameter (Section II D), constrained empirically by the angular-size dependence of $\delta g^{(2)}$ (Section VI). We denote the normalized transverse coherence function by $\Phi(b/r_\perp)$, satisfying $\Phi(0) = 1$ and $\Phi(b/r_\perp) \to 0$ for $b \gg r_\perp$, and treat it as a free function; its model-dependence is discussed below.

Turning to the temporal behavior, quantum foam models posit that each Planck-scale cell of spacetime along the line of sight fluctuates stochastically. Two photons traversing the same cell at coordinate times separated by τ experience path-length fluctuations that are correlated for $\tau \ll \tau_{\rm foam}$ but decorrelated for $\tau \gg \tau_{\rm foam}$, where $\tau_{\rm foam}$ is the characteristic temporal fluctuation timescale at each cell. We introduce $C_\alpha(\tau; D)$ as the normalized *temporal* autocorrelation of the per-cell path-length fluctuation, integrated over the $\sim D/\ell_{\rm P}$ cells along the line of sight (where $\ell_{\rm P} = \sqrt{\hbar G/c^3} \simeq 1.616\times10^{-35}$ m is the Planck length). The timescale $\tau_{\rm foam}$ is an additional free parameter of the foam model, not fixed by the Ng–Perlman variance relation alone. If $\tau_{\rm foam}$ is of order the Planck time, $\ell_{\rm P}/c \approx 5\times10^{-44}$ s, then $C_\alpha(\tau) \approx 0$ for all accessible lags: the foam is fully decorrelated on any observable timescale, the most optimistic scenario for detection. In the opposite *quasi-static* limit $\tau_{\rm foam} \gg D/c$ (foam coherent across the entire line of sight), $C_\alpha(\tau) \approx 1$ for all τ of interest and every temporal observable vanishes.

The Ng–Perlman variance scaling Eq. (1) does not pin the functional forms of $C_\alpha(\tau; D)$ and $\Phi(b/r_\perp)$, and specific foam models predict different shapes. The pixellon model [15] gives a transverse structure with long-range angular components that does not factorize as $\Phi(b/r_\perp)$ (Legendre polynomial series in aperture-plane angle, with the long-range character strongly suppressing image blurring relative to the uncorrelated case); holographic interferometric proposals in the Verlinde–Zurek class [9] tie the metric-fluctuation noise spectrum to the geometry of the propagation region through the holographic principle, implying model-specific temporal correlation structure. We keep C_α and Φ as free functions; quantitative comparison with specific model-predicted kernels would require evaluating the convolution integrals of Sections III–VI with those shapes, which we leave to future work.

The full spatiotemporal covariance between two rays separated by transverse distance b takes the factorized form

$$\left\langle \Delta\ell_{\rm tot}(t;\boldsymbol{\theta})\,\Delta\ell_{\rm tot}(t+\tau;\boldsymbol{\theta}')\right\rangle = \sigma_\ell^2(D)\,C_\alpha(\tau;D)\,\Phi\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right), \tag{4}$$

where $b(\boldsymbol{\theta}, \boldsymbol{\theta}') \simeq D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. The single-ray temporal covariance appropriate for a point source is the special case $\boldsymbol{\theta} = \boldsymbol{\theta}'$ (so $b = 0$):

$$\left\langle \Delta\ell_{\rm tot}(t)\, \Delta\ell_{\rm tot}(t+\tau) \right\rangle = \sigma_\ell^2(D)\, C_\alpha(\tau; D). \tag{5}$$

The product form is a structural assumption of the model, discussed alongside the other approximations in Section II D. The two principal observables developed below — the Siegert discrepancy $\mathcal{S}(\tau)$ and the zero-lag bunching suppression $\delta g^{(2)}$ — both descend from Eq. (1), with Sections III–V probing its longitudinal projection and Section VI its transverse projection.

C. Differential Jitter and Its Distribution

With the spatiotemporal covariance model in place, we define the key derived quantity that enters the coherence integrals. Define the *cross-ray differential jitter* between a photon arriving along direction $\boldsymbol{\theta}$ at time $t + \tau$ and a photon arriving along $\boldsymbol{\theta}'$ at time t:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \equiv \delta t(t+\tau; \boldsymbol{\theta}) - \delta t(t; \boldsymbol{\theta}'). \tag{6}$$

For fixed D, stationarity implies the statistics depend only on τ and the transverse separation $b(\boldsymbol{\theta}, \boldsymbol{\theta}') = D|\boldsymbol{\theta} - \boldsymbol{\theta}'|$. Expanding the square and applying Eq. (4):

$$\begin{aligned}
\sigma_\Delta^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}') &= \frac{1}{c^2} \left\langle [\Delta\ell_{\rm tot}(t+\tau; \boldsymbol{\theta}) - \Delta\ell_{\rm tot}(t; \boldsymbol{\theta}')]^2 \right\rangle \\
&= \frac{1}{c^2} \left[\sigma_\ell^2 + \sigma_\ell^2 - 2\sigma_\ell^2\, C_\alpha(\tau)\, \Phi\!\left(\frac{b}{r_\perp}\right) \right] \\
&= \frac{2\sigma_\ell^2(D)}{c^2} \left[1 - C_\alpha(\tau; D)\, \Phi\!\left(\frac{b(\boldsymbol{\theta}, \boldsymbol{\theta}')}{r_\perp}\right) \right].
\end{aligned} \tag{7}$$

For $b \gg r_\perp$, $\Phi \to 0$ and the variance becomes $2\sigma_\ell^2/c^2$ *at all lags, including* $\tau = 0$ a new effect absent in the point-source case.

By the central limit theorem, $\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau)$ is a sum of contributions from many foam cells and is therefore Gaussian with zero mean:

$$\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}(\tau) \sim \mathcal{N}\big(0,\, \sigma_\Delta^2(\tau; \boldsymbol{\theta}, \boldsymbol{\theta}')\big)\,, \tag{8}$$

where σ_Δ^2 is taken from Eq. (7). Limits of validity of this Gaussian approximation are collected in Section II D.

D. Approximations, Limits, and Free Parameters

The model defined by Eqs. (1), (4), and (8) carries several structural approximations and free parameters. We collect them here so they are not interleaved with the derivations that follow.

Free parameters. The variance coefficient A_α, the temporal correlation $C_\alpha(\tau; D)$ and timescale $\tau_{\rm foam}$, the transverse coherence length $r_\perp$, the transverse coherence function Φ, and (for extended sources) the effective transverse factor $\Phi_{\rm eff}$ introduced in Section VI D parameterize distinct physical effects — single-ray variance, temporal foam dynamics, transverse coherence, and source angular structure — and are not claimed to be simultaneously determined by any single observation.

Dynamical limit. All numerical estimates in Sections III–VI are evaluated in the dynamical limit $C_\alpha \to 0$, which maximizes the predicted signal at any given lag. The computed signals are therefore conditional upper bounds, not predictions of a fully specified foam model.

Factorization of the covariance. The product form of Eq. (4) — separating the total variance from the normalized spatiotemporal shape — is a structural assumption, not a consequence of the Ng–Perlman scaling; in a general stationary process the variance and the normalized correlation functions can carry independent D-dependence. Factorization carries a falsifiable consequence: in the weak-jitter limit $\sigma_\phi \ll 1$, both the point-source Siegert discrepancy $\mathcal{S}(\tau)$ (Section V) and the extended-source bunching suppression $\delta g^{(2)}$ (Section VI E) inherit their entire D-dependence from the single variance factor $\sigma_\phi^2(D) \propto D^{2(1-\alpha)}$. Because $\mathcal{S}(\tau)$ enters linearly in σ_ϕ^2 while $\delta g^{(2)}$ enters linearly in $\sigma_0 \propto \sigma_\phi$ (Section VI E), factorization predicts $\mathcal{S}(\tau) \propto D^{2(1-\alpha)}$ and $\delta g^{(2)} \propto D^{1-\alpha}$ in the fully resolved limit $\Phi_{\rm eff} \to 0$,

where the geometric prefactor $1 - \Phi_{\rm eff} \to 1$ is itself D-independent. For partially resolved sources $\Phi_{\rm eff}$ depends on D only through the dimensionless combination $D\theta_{\rm src}/r_\perp$ (Section VI D), so this combination must be held fixed when distance-scanning a sample. Subject to that condition, sources at a common α distributed in D should trace two power laws whose D-exponents stand in the fixed ratio $2:1$; a measured ratio inconsistent with $2:1$ at fixed α would falsify factorization without requiring a specific non-factorized model.

Gaussianity and CLT breakdown. The Gaussian kernel Eq. (8) relies on the cells contributing to $\Delta_{\boldsymbol{\theta},\boldsymbol{\theta}'}$ being approximately independent. For the cross-ray case the two rays share foam cells along the line of sight, so independence requires $b \gg r_\perp$. In the intermediate regime $b \sim r_\perp$ (equivalently $r_\perp \sim D\Delta\theta$) the Gaussian approximation and the $\Phi_{\rm eff}$ reduction of Section VI D break down together and introduce errors of the same order; they must be assessed jointly.

III. FIRST-ORDER COHERENCE

In Sections III–V we restrict to a point source ($b = 0$), for which the cross-ray differential jitter of Eq. (6) reduces to the single-ray differential jitter

$$\Delta(\tau) \equiv \delta t(t+\tau) - \delta t(t), \tag{9}$$

with variance obtained from Eq. (7) at $\Phi(0) = 1$:

$$\sigma_\Delta^2(\tau; D) = \frac{2\sigma_\ell^2(D)}{c^2}\big[1 - C_\alpha(\tau; D)\big]. \tag{10}$$

A. Jitter Convolution Formula for $g^{(1)}$

The observed first-order coherence is

$$g^{(1)}_{\rm obs}(\tau) = \frac{\langle E^*_{\rm obs}(t)\, E_{\rm obs}(t+\tau)\rangle}{\langle |E_{\rm obs}(t)|^2\rangle}. \tag{11}$$

Substituting Eq. (2) and letting $u = t - \delta t(t)$, $v = t + \tau - \delta t(t+\tau)$, so that $v - u = \tau - \Delta(\tau)$, stationarity of $E_{\rm src}$ gives

$$\langle E^*_{\rm src}(u)\, E_{\rm src}(v)\rangle = \langle |E_{\rm src}|^2\rangle\, g^{(1)}_{\rm src}(\tau - \Delta(\tau)). \tag{12}$$

Since δt is independent of $E_{\rm src}$, averaging over the jitter yields

$$g^{(1)}_{\rm obs}(\tau) = \big\langle g^{(1)}_{\rm src}(\tau - \Delta(\tau))\big\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma_\Delta^2(\tau))\, g^{(1)}_{\rm src}(\tau - \Delta)\, d\Delta. \tag{13}$$

The observed first-order coherence is the intrinsic coherence function convolved with the Gaussian differential-jitter kernel.

B. Closed Forms for Lorentzian and Voigt Sources

As a representative case admitting a closed-form result, we take a Lorentzian line profile, which corresponds to an exponential coherence function via the Wiener–Khinchin theorem:

$$g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_c}, \tag{14}$$

where τ_c is the coherence time. Evaluating Eq. (13) by completing the square and splitting on the sign of $\tau - \Delta$ gives, for $\tau > 0$:

$$g^{(1)}_{\rm obs}(\tau) = e^{\sigma_\Delta^2/2\tau_c^2} \cdot \frac{1}{2}\left[e^{-\tau/\tau_c}\,\mathrm{erfc}\!\left(\frac{\sigma_\Delta/\tau_c - \tau/\sigma_\Delta}{\sqrt{2}}\right) + e^{\tau/\tau_c}\,\mathrm{erfc}\!\left(\frac{\sigma_\Delta/\tau_c + \tau/\sigma_\Delta}{\sqrt{2}}\right)\right]. \tag{15}$$

In the weak-jitter limit $\sigma_\Delta \ll \tau_c$, the Gaussian average of $e^{-|\tau-\Delta|/\tau_c}$ over $\Delta \sim \mathcal{N}(0, \sigma_\Delta^2)$ is dominated by $|\Delta| \ll |\tau|$, so one expands around $\Delta = 0$:

$$g^{(1)}_{\rm obs}(\tau) \approx e^{-|\tau|/\tau_c}\left[1 + \frac{\sigma_\Delta^2(\tau)}{2\tau_c^2} + \mathcal{O}(\sigma_\Delta^4)\right]. \tag{16}$$

Here $\sigma_\Delta \equiv \sigma_\Delta(\tau)$ is τ-dependent through $C_\alpha(\tau)$; the expansion is in $\sigma_\Delta(\tau)/\tau_c$ at fixed τ, and its validity requires $\sigma_\Delta(\tau) \ll \tau_c$ at each lag of interest. The correction is positive because $e^{-|\tau|/\tau_c}$ is convex and Jensen's inequality applies. The normalization $g^{(1)}_{\rm obs}(0) = 1$ is preserved exactly.

For a coherence function $g^{(1)}_{\rm src}(\tau) = e^{-|\tau|/\tau_L}\, e^{-\tau^2/2\tau_G^2}$ whose Wiener–Khinchin transform yields a Voigt line profile — the convolution of a Lorentzian of width $\propto 1/\tau_L$ with a Gaussian of width $\propto 1/\tau_G$ modelling Doppler and turbulent broadening — the jitter kernel combines with the source Gaussian into an effective width and auxiliary parameter,

$$\tilde{\sigma}_1^2 = \frac{\sigma_\Delta^2\,\tau_G^2}{\sigma_\Delta^2 + \tau_G^2}, \qquad \mu_1(\tau) = \frac{\tau_G^2\,\tau}{\sigma_\Delta^2 + \tau_G^2}, \tag{17}$$

giving, for $\tau > 0$:

$$g^{(1)}_{\rm obs}(\tau) = \frac{\tilde{\sigma}_1}{\sigma_\Delta}\, e^{-\tau^2/2(\sigma_\Delta^2+\tau_G^2)}\, e^{\tilde{\sigma}_1^2/2\tau_L^2}\, \frac{1}{2}\left[e^{-\mu_1/\tau_L}\,\mathrm{erfc}\!\left(\frac{\tilde{\sigma}_1/\tau_L - \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right) + e^{\mu_1/\tau_L}\,\mathrm{erfc}\!\left(\frac{\tilde{\sigma}_1/\tau_L + \mu_1/\tilde{\sigma}_1}{\sqrt{2}}\right)\right]. \tag{18}$$

The normalization $g^{(1)}_{\rm obs}(0) = 1$ is preserved: as $\sigma_\Delta \to 0$, $\tilde{\sigma}_1/\sigma_\Delta \to 1$, the exponential prefactors approach unity, and the erfc combination gives 1. In the limit $\tau_G \to \infty$ (pure Lorentzian), $\tilde{\sigma}_1 \to \sigma_\Delta$ and $\mu_1 \to \tau$, so Eq. (18) reduces to Eq. (15). In the limit $\tau_L \to \infty$ (pure Gaussian), the erfc terms combine to unity and $g^{(1)}_{\rm obs}(\tau) = (\tilde{\sigma}_1/\sigma_\Delta)\, e^{-\tau^2/2(\sigma_\Delta^2+\tau_G^2)}$: the observed coherence is a Gaussian whose width is broadened in quadrature by the jitter.

C. Narrow-Band Limit

For a quasi-monochromatic source with center frequency ω_0 and angular-frequency bandwidth $\Delta\omega \ll \omega_0$, write $E_{\rm src}(t) = A(t)\, e^{-i\omega_0 t}$ where $A(t)$ varies on the coherence timescale $\tau_c = 1/\Delta\omega = \lambda^2/(2\pi c\,\Delta\lambda)$. When $\sigma_\ell/(c\tau_c) \ll 1$, typical realizations of δt leave the envelope unaffected, $A(t-\delta t) \approx A(t)$, and the jitter acts purely as a carrier phase modulation. Intuitively, the slowly varying envelope $A(t)$ changes on the much longer coherence timescale τ_c and so is essentially unmoved by a shift $\delta t \ll \tau_c$, while the rapidly oscillating carrier $e^{-i\omega_0 t}$ rotates by $\omega_0\delta t$ — a phase that can be of order unity even when $\delta t/\tau_c$ is negligibly small. The observed field therefore takes the form

$$E_{\rm obs}(t) \approx A(t)\, e^{-i\omega_0 t}\, e^{i\omega_0 \delta t(t)}. \tag{19}$$

Corrections to this envelope approximation are of order $(\sigma_\ell/c\tau_c)^2$: ≈ 0.28 for $\alpha = 1/2$ at the fiducial parameters of Table I, placing the random-walk case at the edge of validity (where the exact convolution Eq. (15) should be used instead), and $\sim 10^{-11}$ for $\alpha = 2/3$, where the narrow-band derivation below is exact to any practical precision. Since $A(t)$ and $\delta t(t)$ are independent, the first-order coherence factorizes. The jitter contribution is the characteristic function of $\Delta(\tau)$ evaluated at ω_0; since $\Delta(\tau)$ is Gaussian (Section II C),

$$\left\langle e^{i\omega_0 \Delta(\tau)} \right\rangle_{\delta t} = e^{-\omega_0^2 \sigma_\Delta^2(\tau)/2}. \tag{20}$$

This holds when $\tau \gg \tau_{\rm foam}$ so that foam-cell contributions are independent; in the dynamical limit $\tau_{\rm foam} \ll \tau_c$ the condition is satisfied at all observationally relevant lags. Substituting $\omega_0 = 2\pi c/\lambda$ and Eq. (10):

$$g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \exp\!\left(-\sigma_\phi^2\left[1 - C_\alpha(\tau; D)\right]\right), \tag{21}$$

where we define the rms accumulated phase

$$\sigma_\phi \equiv \frac{2\pi\,\sigma_\ell(D)}{\lambda} = \frac{2\pi\,\sqrt{A_\alpha}\,D^{1-\alpha}\,\ell_{\rm P}^{\alpha}}{\lambda}. \tag{22}$$

We write the foam suppression factor as

$$\mathcal{F}_{\rm foam}(\tau) \equiv e^{-\sigma_\phi^2[1 - C_\alpha(\tau;D)]}, \tag{23}$$

so $g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau)$. At $\tau = 0$: $\mathcal{F}_{\rm foam}(0) = 1$. At $\tau \to \infty$: $\mathcal{F}_{\rm foam}(\infty) = e^{-\sigma_\phi^2[1 - C_\alpha(\infty;D)]}$, which equals $e^{-\sigma_\phi^2}$ in the dynamical limit ($C_\alpha \to 0$) and approaches unity for quasi-static foam ($C_\alpha \approx 1$).

D. Regime Summary

The exact convolution Eq. (13) (and its Lorentzian closed form Eq. (15)) is always valid within the model. The weak-jitter expansion Eq. (16) requires $\sigma_\Delta(\tau)/\tau_c \ll 1$; the narrow-band carrier-phase result Eq. (21) requires $\sigma_\ell/(c\tau_c) \ll 1$ and captures the dominant physics when $\sigma_\phi \gtrsim 1$ but $\sigma_\phi \ll \lambda/\Delta\lambda$. The two conditions are compatible and govern distinct effects (envelope displacement vs. carrier-phase suppression); when neither holds, Eq. (13) must be evaluated directly.

E. Connection to Fringe Visibility and Strehl Ratio

The fringe visibility and Strehl ratio are the observables through which published constraints on spacetime foam have been obtained [4, 11–14]. Both are direct consequences of the first-order coherence $g^{(1)}_{\rm obs}(\tau)$ derived above, and the foam suppression factor $\mathcal{F}_{\rm foam}(\tau)$ of Eq. (23) unifies them in a single framework.

1. *Fringe Visibility and the Lieu–Hillman Condition*

The fringe visibility measured in a stellar interferometer at time delay τ is $V(\tau) = |g^{(1)}_{\rm obs}(\tau)|$. From Eq. (21):

$$V(\tau) = V_{\rm src}(\tau) \cdot \mathcal{F}_{\rm foam}(\tau), \tag{24}$$

where $V_{\rm src}(\tau) = |g^{(1)}_{\rm src}(\tau)|$ is the intrinsic source visibility. Lieu & Hillman [11] argue that fringes are erased when the accumulated phase randomizes over $[0, 2\pi)$, i.e. $\sigma_\phi \gtrsim 1$ — the regime in which the rms accumulated phase exceeds one radian and randomizes the phase across the aperture, erasing interferometric fringes. In our language this is the limit $\mathcal{F}_{\rm foam} \to 0$: the foam suppression factor collapses and $V \to 0$. The Lieu–Hillman condition $\Delta\phi \geq 2\pi$, where $\Delta\phi \simeq 2\pi\sigma_\ell(D)/\lambda$ [4], is precisely $\sigma_\phi \geq 1$. Our result extends this binary criterion to a continuous suppression curve $V(\tau)$ whose shape encodes the temporal decorrelation $C_\alpha(\tau; D)$.

2. *Strehl Ratio*

The Strehl ratio S the ratio of the observed PSF peak to the diffraction-limited peak is the standard measure of wavefront quality in telescope optics. In the Ng–Perlman picture [4], spacetime foam imprints a spatially varying random phase $\Delta\phi(x, y)$ across the entrance aperture: each point (x, y) on the mirror receives a ray that has accumulated an independent random path-length shift from the Planck-scale cells it traversed, producing a random phase screen with rms value σ_ϕ. When these phase shifts are spatially uncorrelated across the aperture the limit $r_\perp \ll D_{\rm tel}$, where $r_\perp$ is the transverse foam coherence length and $D_{\rm tel}$ is the aperture diameter the Maréchal approximation (the small-phase-variance expansion of the diffraction-limited PSF reduction) gives

$$S = e^{-\sigma_\phi^2}, \tag{25}$$

valid for $\sigma_\phi \lesssim 1$; for $\sigma_\phi \gg 1$ the PSF peak is effectively destroyed.

The same σ_ϕ^2 appears in the temporal framework as the large-lag limit of $\mathcal{F}_{\rm foam}$. In the dynamical limit $C_\alpha(\infty; D) \to 0$, Eq. (23) gives

$$S = \mathcal{F}_{\rm foam}(\tau \to \infty) = e^{-\sigma_\phi^2}, \tag{26}$$

so the spatial and temporal descriptions reduce to the same Gaussian characteristic function: summing independent phase contributions across the aperture at a fixed time, or tracking the phase along a single ray as the lag grows large enough that the two time samples become independent. For quasi-static foam ($C_\alpha \approx 1$) the temporal result instead gives $\mathcal{F}_{\rm foam}(\infty) \to 1$, consistent with $S \to 1$ in the $r_\perp \gg D_{\rm tel}$ limit.

The two channels do, however, depend differently on $r_\perp$. The Marchal formula (25) requires adjacent aperture elements to sample *independent* foam realizations, i.e. $r_\perp \ll D_{\rm tel}$; in the opposite limit $r_\perp \gg D_{\rm tel}$ the wavefront acquires a common time-varying phase offset with no spatial variation and $S \to 1$ even when σ_ϕ^2 is large. The temporal $g^{(1)}$ of Eq. (21), by contrast, probes the phase along a single ray as a function of lag and is insensitive to $r_\perp$ relative to the aperture. The transverse coherence function $\Phi(b/r_\perp)$ governs a physically distinct effect — correlations between rays from different sky directions — which manifests in the second-order coherence of extended sources (Section VI).

IV. SECOND-ORDER COHERENCE

We derive the second-order coherence $g^{(2)}_{\rm obs}(\tau)$ for a point source, continuing the framework of Section III. The result has the same structure as Eq. (13): $g^{(2)}_{\rm obs} - 1$ is a convolution of $g^{(2)}_{\rm src} - 1$ with the identical Gaussian jitter kernel. Because the intensity $|E_{\rm obs}|^2$ is blind to the carrier phase $e^{i\omega_0\delta t}$, foam leaves $g^{(2)}$ nearly unchanged in the narrow-band limit while strongly suppressing $g^{(1)}$ — the asymmetry that produces the Siegert violation developed in Section V.

A. Jitter Convolution Formula for $g^{(2)}$

Fix a realization of the jitter process $\delta t(\cdot)$. Conditioned on it, $E_{\rm obs}(t)$ is a deterministic time-warp of a complex Gaussian field, so the Siegert identity holds at each fixed pair $(t, t+\tau)$ as a purely algebraic Gaussian moment identity, with no stationarity required. (Ergodicity of $E_{\rm src}$ allows the conditional time-average to be replaced by the conditional ensemble average; the subsequent average over the stationary δt process then yields a function of τ only, by independence of $E_{\rm src}$ and δt.) With $g^{(1)}_{\rm obs}(\tau \mid \delta t) = g^{(1)}_{\rm src}(\tau - \Delta(\tau))$ and the intrinsic Siegert relation $g^{(2)}_{\rm src} - 1 = |g^{(1)}_{\rm src}|^2$:

$$g^{(2)}_{\rm obs}(\tau \mid \delta t) - 1 = g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1. \tag{27}$$

Averaging over the jitter distribution:

$$g^{(2)}_{\rm obs}(\tau) - 1 = \left\langle g^{(2)}_{\rm src}(\tau - \Delta(\tau)) - 1 \right\rangle_{\delta t} = \int_{-\infty}^{\infty} \mathcal{N}(\Delta; 0, \sigma^2_\Delta(\tau)) \left[g^{(2)}_{\rm src}(\tau - \Delta) - 1 \right] d\Delta. \tag{28}$$

This Gaussian kernel relies on the CLT argument of Section II C, which requires $\tau \gg \tau_{\rm foam}$; for $\tau \lesssim \tau_{\rm foam}$ the kernel may deviate from Gaussian and Eq. (28) should be regarded as approximate in that regime (see Section II B for the definition and limits of $\tau_{\rm foam}$).

B. Narrow-Band Limit: $g^{(2)}$ is Unchanged to Leading Order

The intensity $I_{\rm obs}(t) = |E_{\rm obs}(t)|^2$ is blind to the carrier phase $e^{i\omega_0\delta t(t)}$: only the envelope matters. In the narrow-band limit $\delta t \ll \tau_c$, the jitter does not appreciably shift the envelope, $A(t - \delta t) \approx A(t)$, so $I_{\rm obs}(t) \approx |A(t)|^2 = I_{\rm src}(t)$ to leading order in $\delta t/\tau_c$. Consequently:

$$g^{(2)}_{\rm obs}(\tau) = g^{(2)}_{\rm src}(\tau) \qquad \text{(leading order in } \delta t/\tau_c), \tag{29}$$

with corrections of order $(\sigma_\ell/c\tau_c)^2$ from both envelope-timing and carrier-phase contributions to the intensity, of the same magnitude as those discussed for $g^{(1)}$ in Section III C: at the $\sim$8% level for the (already excluded) $\alpha = 1/2$ benchmark and $\sim 10^{-11}$ for $\alpha = 2/3$.

By contrast, $g^{(1)}$ *is* sensitive to the carrier phase because $\sigma_\phi = \omega_0\sigma_\ell/c$ can be large even when $\delta t \ll \tau_c$: at the same fiducial parameters, $\sigma_\phi \approx 280 \gg 1$.

V. SIEGERT RELATION VIOLATION

The Siegert relation $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$ holds for any stationary complex Gaussian field and provides a powerful internal consistency check: once $g^{(1)}$ is known, the expected $g^{(2)}$ is fixed. As foreshadowed in Section IV, foam violates this relation because $g^{(1)}_{\rm obs}$ is suppressed by the carrier-phase factor $\mathcal{F}_{\rm foam}$ while $g^{(2)}_{\rm obs}$ is not. This section derives and characterizes the Siegert discrepancy $\mathcal{S}(\tau) \equiv [g^{(2)}_{\rm obs} - 1] - |g^{(1)}_{\rm obs}|^2$.

A. General Bound

Define the Siegert discrepancy:

$$\mathcal{S}(\tau) \equiv \left[g^{(2)}_{\rm obs}(\tau) - 1 \right] - \left| g^{(1)}_{\rm obs}(\tau) \right|^2. \tag{30}$$

Let $X \equiv g^{(1)}_{\rm src}(\tau - \Delta(\tau))$, a complex random variable. From Eqs. (13) and (28):

$$g^{(1)}_{\rm obs}(\tau) = \langle X \rangle_{\delta t}, \qquad g^{(2)}_{\rm obs}(\tau) - 1 = \langle |X|^2 \rangle_{\delta t}. \tag{31}$$

Therefore

$$\mathcal{S}(\tau) = \langle |X|^2 \rangle - |\langle X \rangle|^2 \geq 0, \tag{32}$$

by the Cauchy–Schwarz inequality, with equality only when $\Delta(\tau)$ has zero variance (i.e., at $\tau = 0$ or in the absence of foam). The Siegert relation is violated for all $\tau \neq 0$ whenever foam is present. This non-negativity holds for a point source; for extended sources the cross-ray differential jitter can drive $\mathcal{S}(0) < 0$ (see Section VI). The Siegert violation is not a non-classical signature: the foam-modified field remains a classical stochastic process, and the classical inequalities $g^{(2)}(0) \geq g^{(2)}(\tau)$ and $g^{(2)}(0) > 1$ are preserved; it is only the Gaussian-statistics identity relating $g^{(2)}$ to $g^{(1)}$ that is broken.

B. Sensitivity Scale Decoupling

The violation arises because $g^{(1)}$ and $g^{(2)}$ respond to foam on different scales. The first-order coherence is controlled by $\sigma_\phi = 2\pi\sigma_\ell(D)/\lambda$, the rms accumulated phase along a single ray; $g^{(1)}$ is sensitive to it because the carrier phase $e^{i\omega_0 \delta t}$ appears directly in the field correlation. The second-order coherence is controlled by $\sigma_\Delta(\tau)/\tau_c$, the rms differential jitter scaled by the coherence time; $g^{(2)}$ is sensitive to it because intensity correlations depend on how much the jitter *changes* between two detection events, not its absolute value. These two scales are related by

$$\frac{\sigma_\Delta(\tau)}{\tau_c} = \frac{\sqrt{2[1 - C_\alpha(\tau)]}\,\sigma_\ell(D)}{c\tau_c} = \sigma_\phi \cdot \frac{\Delta\lambda}{\lambda} \cdot \sqrt{2[1 - C_\alpha(\tau)]}, \tag{33}$$

using $\omega_0 \tau_c = \lambda/\Delta\lambda$. The factor $\Delta\lambda/\lambda \ll 1$ means that even when $\sigma_\phi \gg 1$ (fringes erased), $\sigma_\Delta(\tau)/\tau_c$ can be small so $g^{(2)}$ remains near its intrinsic value while $g^{(1)}$ is suppressed. This decoupling is the physical root of the Siegert violation, and is only possible within the narrow-band regime discussed in Section III C. Numerical values of both scales are given in Table I (evaluated at the dynamical limit $C_\alpha \to 0$; for quasi-static foam $C_\alpha \approx 1$ the ratio vanishes at all lags); the corresponding foam-amplitude curves are shown against the principal sensitivity thresholds in Fig. 1.

TABLE I. Sensitivity scales at $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$, $C_\alpha \to 0$. The $\delta g^{(2)}$ row is evaluated in the resolved limit $\Phi_{\rm eff} \to 0$ for a Lorentzian source. The $\alpha = 1/2$ column is observationally excluded [14] and at the edge of narrow-band validity; the $\alpha = 0.55$ column illustrates an in-domain case viable for foam models with long-range transverse correlations [15]; $\alpha = 2/3$ is the holographic case.

Scale	$\alpha = 1/2$ (excluded)	$\alpha = 0.55$ (in-domain)	$\alpha = 2/3$
σ_ϕ	280	0.27	2.5×10^{-8}
$\sigma_\phi\,\Delta\lambda/\lambda$	0.28 (marginal)	2.7×10^{-4}	2.5×10^{-11}
$\sigma_\Delta(\tau)/\tau_c$	0.40	3.8×10^{-4}	3.6×10^{-11}
$\delta g^{(2)}$	— (nonlinear)	6.1×10^{-4}	5.7×10^{-11}

C. Explicit Form and Properties in the Narrow-Band Limit

In the narrow-band limit, $g^{(2)}_{\rm obs} \approx g^{(2)}_{\rm src}$ (Eq. (29)), while $g^{(1)}_{\rm obs}$ is suppressed by $\mathcal{F}_{\rm foam}(\tau)$ (Eq. (21)). Using the intrinsic Siegert relation $g^{(2)}_{\rm src}(\tau) - 1 = |g^{(1)}_{\rm src}(\tau)|^2$, to leading order:

$$\mathcal{S}(\tau) \approx |g^{(1)}_{\rm src}(\tau)|^2 \Big(1 - e^{-2\sigma_\phi^2[1 - C_\alpha(\tau;D)]}\Big). \tag{34}$$

In the weak-jitter limit $\sigma_\phi \ll 1$, $\mathcal{S} \approx 2\sigma_\phi^2[1 - C_\alpha]\,|g^{(1)}_{\rm src}|^2$ linear in σ_ϕ^2 and small. In the strong-jitter limit $\sigma_\phi \gg 1$, the exponential vanishes and $\mathcal{S} \approx |g^{(1)}_{\rm src}(\tau)|^2$: $g^{(2)}_{\rm obs} - 1$ retains its intrinsic value while $|g^{(1)}_{\rm obs}|^2 \to 0$, producing a full violation. Quasi-static foam ($C_\alpha \approx 1$) gives $\mathcal{S} \approx 0$ at all lags foam is invisible to all coherence observables in this limit. The discrepancy $\mathcal{S}(\tau)$ is a direct observable both $g^{(1)}_{\rm obs}$ and $g^{(2)}_{\rm obs}$ are measurable from the same dataset and the modulation beyond the prefactor $|g^{(1)}_{\rm src}(\tau)|^2$ is controlled solely by α, D, and $C_\alpha(\tau; D)$, independently of the unknown source spectrum.

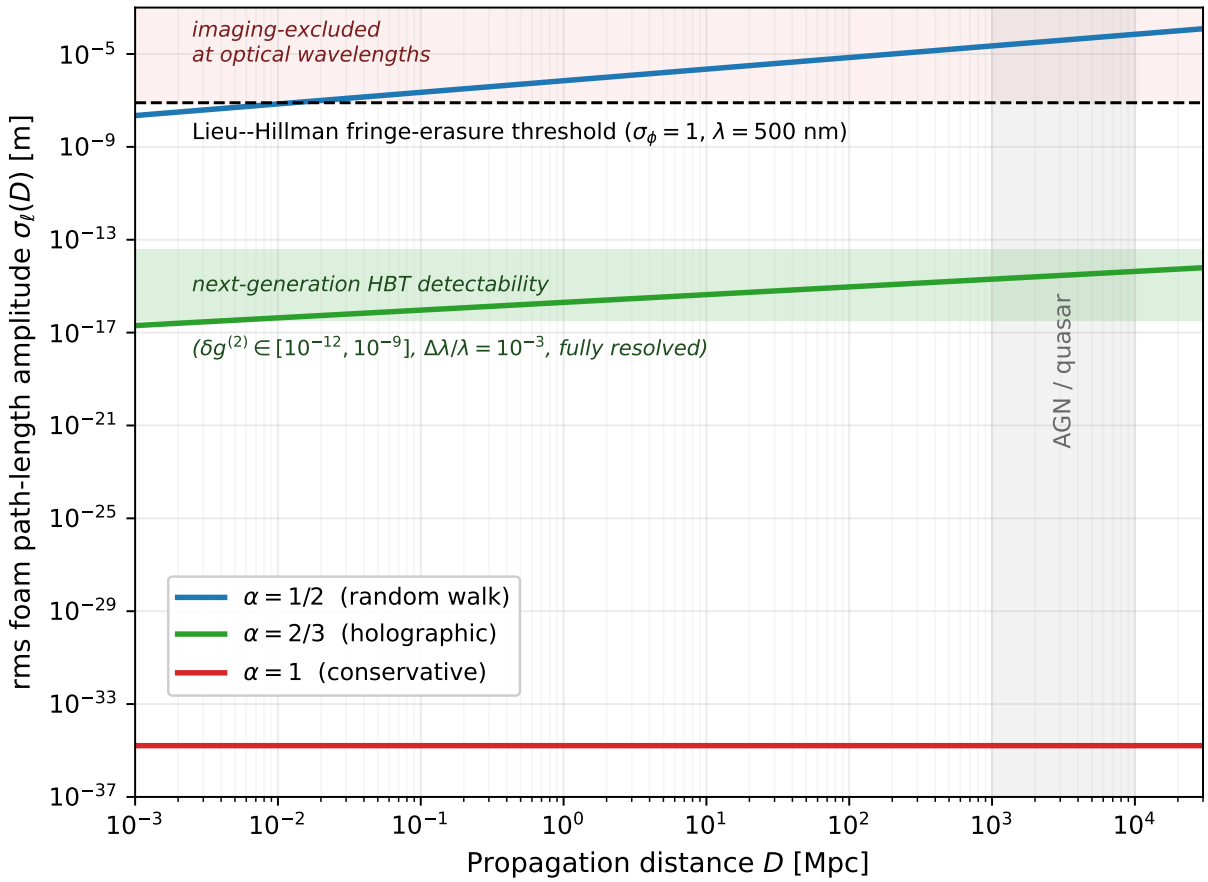

FIG. 1. Foam-induced rms path-length amplitude $\sigma_\ell(D)$ for the three Ng–Perlman exponents (taking $A_\alpha = 1$ in the path-length variance Eq. (1)), shown against the principal optical sensitivity thresholds. The horizontal dashed line is the Lieu–Hillman fringe-erasure threshold $\sigma_\ell = \lambda/(2\pi)$ at $\lambda = 500$ nm; the shaded region above it is excluded by extragalactic imaging tests [11–14]. The shaded green band is the next-generation HBT detectability range at fiducial bandwidth $\Delta\lambda/\lambda = 10^{-3}$ in the fully resolved limit ($\Phi_{\rm eff} \to 0$), corresponding to $\delta g^{(2)} \in [10^{-12}, 10^{-9}]$ between the modal shot-noise floor and the projected systematic floor. The vertical grey band marks the AGN/quasar redshift range $D \sim 1$–10 Gpc. The random-walk exponent $\alpha = 1/2$ crosses the imaging exclusion at $D \sim 13$ kpc and is therefore ruled out for any extragalactic target. The holographic exponent $\alpha = 2/3$ sits within the HBT detectability band over the AGN regime, indicating that the second-order observables developed in Sections V–VI approach the boundary of next-generation HBT instrumentation. The conservative exponent $\alpha = 1$ has $\sigma_\ell = \ell_{\rm P}$ regardless of D and lies far below all current and projected sensitivity floors.

VI. EXTENDED SOURCES: TRANSVERSE FOAM COHERENCE

We now generalize from a point source to an extended incoherent source. The first-order coherence $g^{(1)}_{\rm obs}$ turns out to be unchanged from the point-source result, because $g^{(1)}$ probes only single-ray statistics. The second-order coherence $g^{(2)}_{\rm obs}$, however, acquires new structure through cross-ray differential jitter between rays arriving from different sky directions, producing a suppression of the zero-lag bunching peak that directly probes the transverse foam coherence length $r_\perp$.

A. Field Model and Spatiotemporal Covariance

We generalize to an extended incoherent source whose angular brightness distribution is $I(\boldsymbol{\theta})$ (normalized to unity). The analysis assumes a point (single spatial mode) detector; a finite aperture would introduce Van Cittert–Zernike geometric phases that modify the angular coherence. The detected field is a weighted superposition over rays labelled

by direction offset $\boldsymbol{\theta}$:

$$E_{\rm obs}(t) = \int d\boldsymbol{\theta}\, I^{1/2}(\boldsymbol{\theta})\, E_{\boldsymbol{\theta}}(t - \delta t(t;\boldsymbol{\theta}))\,, \tag{35}$$

where the $E_{\boldsymbol{\theta}}$ are independent, identically distributed complex Gaussian processes (one per sky direction), and each ray carries its own spacetime foam jitter $\delta t(t;\boldsymbol{\theta})$. The spatiotemporal covariance of the path-length aggregate is given by Eq. (4) of Section II, with $b(\boldsymbol{\theta},\boldsymbol{\theta}') \simeq D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$ the transverse separation between rays evaluated at the source end. For foam distributed uniformly along the line of sight the ray separation at position z is $z|\boldsymbol{\theta}-\boldsymbol{\theta}'|$, so the full cross-covariance integral involves an average over $0 \le z \le D$; setting $b = D|\boldsymbol{\theta}-\boldsymbol{\theta}'|$ evaluates the separation at the source plane. Because $r_\perp$ is treated as an effective free parameter, the $O(1)$ geometric factor from the z-integral is absorbed into its definition; its value is constrained empirically by the angular-size dependence of $\delta g^{(2)}$, derived in Section VI D.

B. First-Order Coherence is Unchanged

The numerator of $g^{(1)}_{\rm obs}(\tau)$ involves cross terms $\langle E^*_{\boldsymbol{\theta}}(\ldots)\, E_{\boldsymbol{\theta}'}(\ldots)\rangle$. Because different source directions are spatially incoherent,

$$\langle E^*_{\boldsymbol{\theta}}(s)\, E_{\boldsymbol{\theta}'}(s')\rangle \propto \delta(\boldsymbol{\theta}-\boldsymbol{\theta}'), \tag{36}$$

so only the diagonal $\boldsymbol{\theta}=\boldsymbol{\theta}'$ contributes. Each diagonal term reproduces the single-ray result of Section III C, and the angular integral over I gives unity. Therefore:

$$g^{(1)}_{\rm obs}(\tau) = g^{(1)}_{\rm src}(\tau)\cdot e^{-\sigma^2_\phi[1-C_\alpha(\tau;D)]}, \tag{37}$$

identical to the point-source result Eq. (21). The first-order coherence probes only single-ray statistics; the transverse foam structure leaves no imprint on $g^{(1)}$.

C. Second-Order Coherence for an Extended Source

The unnormalized intensity correlator is $\langle I(t)I(t+\tau)\rangle = \langle E^*(t)E(t)E^*(t+\tau)E(t+\tau)\rangle$, where $E \equiv E_{\rm obs}$ is the field of Eq. (35). Expanding the double angular integral and applying the Wick (moment) theorem to the four-point function of the zero-mean complex Gaussian field $E_{\boldsymbol{\theta}}$ produces two pairings (the $I^{1/2}$ weight factors enter the angular integral and are suppressed here for clarity):

$$\langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_2} E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_4}\rangle = \langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_2}\rangle\langle E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_4}\rangle + \langle E^*_{\boldsymbol{\theta}_1} E_{\boldsymbol{\theta}_4}\rangle\langle E^*_{\boldsymbol{\theta}_3} E_{\boldsymbol{\theta}_2}\rangle. \tag{38}$$

The first pairing contributes $\langle I(t)\rangle\langle I(t+\tau)\rangle$ (the disconnected term, which cancels when forming $g^{(2)}-1$). The second pairing gives the connected contribution. Because different source directions are spatially incoherent, $\langle E^*_{\boldsymbol{\theta}}(s)\, E_{\boldsymbol{\theta}'}(s')\rangle \propto \delta(\boldsymbol{\theta}-\boldsymbol{\theta}')$, so the connected pairing requires $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_4$ and $\boldsymbol{\theta}_3 = \boldsymbol{\theta}_2$. The four-fold angular integral therefore collapses to a double integral over ray pairs $(\boldsymbol{\theta},\boldsymbol{\theta}')$, each pair contributing its own jitter-convolved $g^{(2)}_{\rm src}$:

$$g^{(2)}_{\rm obs}(\tau) - 1 = \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; I(\boldsymbol{\theta})I(\boldsymbol{\theta}') \int_{-\infty}^{\infty} \mathcal{N}\big(\Delta; 0, \sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')\big)\; \big[g^{(2)}_{\rm src}(\tau-\Delta) - 1\big]\, d\Delta, \tag{39}$$

where $\sigma^2_\Delta(\tau;\boldsymbol{\theta},\boldsymbol{\theta}')$ is from Eq. (7) (Section II C). The key new feature relative to the point-source case is that for $b \gg r_\perp$, $\Phi \to 0$ and $\sigma^2_\Delta \to 2\sigma^2_\ell/c^2$ at all lags, including $\tau = 0$. In the limit $\Phi \equiv 1$ (all rays transversely coherent), every kernel reduces to the single-ray $\mathcal{N}(0, \sigma^2_\Delta(\tau))$ and Eq. (39) reduces to the point-source result Eq. (28).

D. Effective Transverse Coherence Factor

To reduce Eq. (39) to a tractable form, define the effective transverse coherence factor:

$$\Phi_{\rm eff} \equiv \iint d\boldsymbol{\theta}\, d\boldsymbol{\theta}'\; I(\boldsymbol{\theta})I(\boldsymbol{\theta})\, \Phi\!\left(\frac{b(\boldsymbol{\theta},\boldsymbol{\theta}')}{r_\perp}\right). \tag{40}$$

This replaces the ray-dependent differential variance with a single effective variance:

$$\sigma^2_{\tau,\mathrm{eff}}(\tau) \equiv \frac{2\sigma^2_\ell(D)}{c^2}\Big[1 - C_\alpha(\tau; D)\,\Phi_{\mathrm{eff}}\Big], \tag{41}$$

and allows Eq. (39) to be approximated by the simpler 1D convolution of Eq. (28) with σ^2_Δ replaced by $\sigma^2_{\tau,\mathrm{eff}}$. The two limits are: $\Phi_{\mathrm{eff}} \to 1$ (point-like source or $r_\perp \gg D\,\Delta\theta$) recovers the single-ray result; $\Phi_{\mathrm{eff}} \to 0$ (fully incoherent, $r_\perp \ll D\,\Delta\theta$) maximizes the effective variance. The approximation becomes exact in both limits; the relative error of the single-kernel approximation is bounded above by $(1-\Phi_{\mathrm{eff}})/\Phi_{\mathrm{eff}}$, peaking at $\sim 50\%$ near $\Phi_{\mathrm{eff}} = 1/2$ ($D\theta_{\mathrm{src}} \sim r_\perp$) where the cross-ray Gaussian approximation also degrades (Section II D). For the resolved-limit applications of Section VI F, $\Phi_{\mathrm{eff}} \ll 1$ and the single-kernel approximation is controlled.

E. Extended-Source Siegert Discrepancy and Zero-Lag Bunching

For the extended source, $g^{(1)}_{\mathrm{obs}}(\tau)$ is unchanged from the point-source result (Eq. 37), while $g^{(2)}_{\mathrm{obs}}(\tau)$ is given by the general mixture Eq. (39). Unlike the point-source case, in the narrow-band limit $g^{(2)}_{\mathrm{obs}}$ is *not* equal to $g^{(2)}_{\mathrm{src}}$: the cross-ray differential jitter modifies the intensity correlation through the envelope timing, even though the carrier phase does not affect individual intensities. The Siegert discrepancy is therefore

$$\mathcal{S}(\tau) = \Big[g^{(2)}_{\mathrm{obs}}(\tau) - 1\Big] - \Big|g^{(1)}_{\mathrm{obs}}(\tau)\Big|^2, \tag{42}$$

where $g^{(2)}_{\mathrm{obs}}(\tau) - 1$ is given by Eq. (39) and $|g^{(1)}_{\mathrm{obs}}(\tau)|^2 = |g^{(1)}_{\mathrm{src}}(\tau)|^2\, e^{-2\sigma^2_\phi[1-C_\alpha(\tau)]}$.

At $\tau = 0$, $g^{(1)}_{\mathrm{obs}}(0) = 1$ exactly by the spatial-incoherence argument of Section VI, so $\mathcal{S}(0) = g^{(2)}_{\mathrm{obs}}(0) - 2$. Evaluating the zero-lag effective variance with $C_\alpha(0; D) = 1$ gives

$$\sigma^2_0 \equiv \frac{2\sigma^2_\ell(D)}{c^2}(1 - \Phi_{\mathrm{eff}}), \tag{43}$$

which is strictly positive whenever $\Phi_{\mathrm{eff}} < 1$, so the ray-pair mixture reduces to a single Gaussian kernel of width σ_0.

For a Lorentzian source, $g^{(1)}_{\mathrm{src}}(\tau) = e^{-|\tau|/\tau_c}$, so the Siegert relation gives $g^{(2)}_{\mathrm{src}}(\tau) - 1 = e^{-2|\tau|/\tau_c}$. The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma^2_0)$ then becomes, setting $s \equiv \sigma_0/\tau_c$ and splitting on the sign of Δ (both halves contributing equally by symmetry of $|\Delta|$),

$$\big\langle e^{-2|\Delta|/\tau_c}\big\rangle = \frac{2}{\sqrt{2\pi}\,\sigma_0}\int_0^\infty e^{-2\Delta/\tau_c} e^{-\Delta^2/2\sigma_0^2} d\Delta = e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \tag{44}$$

which gives 1 as $s \to 0$ (no suppression) and $\to 0$ as $s \to \infty$ via the standard asymptotic $e^{x^2}\mathrm{erfc}(x) \sim 1/(x\sqrt{\pi})$ (full suppression, $g^{(2)}_{\mathrm{obs}}(0) \to 1$). So

$$g^{(2)}_{\mathrm{obs}}(0) = 1 + e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \qquad s \equiv \sigma_0/\tau_c, \tag{45}$$

and the zero-lag bunching peak is suppressed below its ideal value of 2 by

$$\delta g^{(2)} \equiv 2 - g^{(2)}_{\mathrm{obs}}(0) = 1 - e^{2s^2}\,\mathrm{erfc}\Big(\sqrt{2}\,s\Big), \tag{46}$$

giving

$$\mathcal{S}(0) = -\,\delta g^{(2)} \le 0. \tag{47}$$

This sign is qualitatively distinct from the point-source result $\mathcal{S}(\tau \neq 0) \ge 0$ of Section V: a negative zero-lag Siegert discrepancy is therefore a sign-flip diagnostic that distinguishes spatially resolved from unresolved foam signatures. In the point-source limit $\Phi_{\mathrm{eff}} \to 1$, $\sigma_0 \to 0$, $\delta g^{(2)} \to 0$, and $\mathcal{S}(0) \to 0$, recovering the point-source result. Since $\sigma^2_0 \propto (1 - \Phi_{\mathrm{eff}})$ and Φ_{eff} decreases as the source angular diameter increases at fixed D and $r_\perp$, more extended sources experience greater suppression of the bunching peak and a more negative Siegert discrepancy at zero lag.

For the astrophysically more representative Voigt-derived $g^{(1)}_{\mathrm{src}}(\tau) = e^{-|\tau|/\tau_L}\, e^{-\tau^2/2\tau_G^2}$ introduced in Section III (Lorentzian of width τ_L convolved with a Gaussian of width τ_G, modelling natural plus Doppler/turbulent broadening),

the intrinsic Siegert relation gives $g^{(2)}_{\rm src}(\tau) - 1 = e^{-2|\tau|/\tau_L}\, e^{-\tau^2/\tau_G^2}$. The zero-lag convolution over $\Delta \sim \mathcal{N}(0, \sigma_0^2)$ has the same structure as the first-order Voigt convolution of Eqs. (17)–(18), with $\sigma_\Delta \to \sigma_0$ and the Lorentzian width halved ($\tau_L \to \tau_L/2$ inside the kernel, equivalently the factor of 2 in $2\sigma_0^2$ below). The effective width is

$$\tilde{\sigma}_2^2 = \frac{\sigma_0^2\, \tau_G^2}{2\sigma_0^2 + \tau_G^2}, \tag{48}$$

yielding

$$g^{(2)}_{\rm obs}(0) - 1 = \frac{\tilde{\sigma}_2}{\sigma_0}\, e^{2\tilde{s}_2^2}\, \mathrm{erfc}\Big(\sqrt{2}\, \tilde{s}_2\Big), \qquad \tilde{s}_2 \equiv \tilde{\sigma}_2/\tau_L. \tag{49}$$

In the limit $\tau_G \to \infty$ (pure Lorentzian), $\tilde{\sigma}_2 \to \sigma_0$ and Eq. (49) reduces to Eq. (45). In the limit $\tau_L \to \infty$ (pure Gaussian), $\tilde{s}_2 \to 0$ and $g^{(2)}_{\rm obs}(0) - 1 = \tilde{\sigma}_2/\sigma_0 = \tau_G/\sqrt{2\sigma_0^2 + \tau_G^2}$, with no $O(s)$ correction.

In the narrow-band limit $\sigma_\ell/(c\tau_c) \ll 1$ (Section III C), $\sigma_0 \ll \tau_c$ for any $\Phi_{\rm eff}$. Substituting $u = \Delta/\sigma_0$ and expanding $g^{(2)}_{\rm src}(\sigma_0 u) - 1$ for small σ_0, the convolution integral yields to first order in $s = \sigma_0/\tau_c$:

$$g^{(2)}_{\rm obs}(0) - 1 \approx \big[g^{(2)}_{\rm src}(0) - 1\big] + \sqrt{\tfrac{2}{\pi}}\, \sigma_0\, g^{(2)\prime}_{\rm src}(0^+), \tag{50}$$

where $g^{(2)\prime}_{\rm src}(0^+)$ denotes the right-derivative of $g^{(2)}_{\rm src}$ at zero lag (the function is non-differentiable at $\tau = 0$ for cusped profiles such as the Lorentzian, so the symmetric derivative does not exist there). For sources with a smooth (even) profile the right-derivative vanishes and the leading correction is $O(\sigma_0^2)$; for a Lorentzian, $g^{(2)\prime}_{\rm src}(0^+) = -2/\tau_c$, recovering $\delta g^{(2)} \approx 2\sqrt{2/\pi}\, s$, consistent with the small-s expansion of Eq. (46). For a Voigt profile, the same expansion gives $\delta g^{(2)} \approx 2\sqrt{2/\pi}\, \sigma_0/\tau_L + (\sigma_0/\tau_G)^2 + \cdots$: the O(s) term is controlled solely by the Lorentzian component τ_L, and Gaussian broadening (Doppler or instrumental) enters only at O(s^2). The O(s) scaling therefore survives whenever the line has any Lorentzian component, provided the line is spectrally resolved at the scale of τ_L. This is consistent with the sensitivity-scale hierarchy of Section V B: the zero-lag suppression is small in the narrow-band regime. Observationally, this implies that a spectrograph transmitting a nearly flat bandpass (e.g. continuum), whose $g^{(2)}_{\rm src}$ is smooth at zero lag, suppresses the first-order contribution, leaving only an $O(s^2)$ residual. Maximizing foam sensitivity in $\delta g^{(2)}$ therefore favours emission lines with a significant Lorentzian component (natural or pressure broadening) that is spectrally resolved by the instrument; if the line is unresolved, the observed profile is the instrumental response (approximately Gaussian), the cusp is erased, and the leading correction reverts to O(s^2).

The suppression $\delta g^{(2)}$ arises from a purely spatial mechanism with no analogue in the point-source case: even at $\tau = 0$, photon pairs from different sky directions $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$ accumulate independent instantaneous jitters $\delta t(t; \boldsymbol{\theta}) \neq \delta t(t; \boldsymbol{\theta}')$ when $b \gg r_\perp$, and their relative offset blurs the $g^{(2)}(0)$ measurement. Because $|g^{(1)}_{\rm obs}(0)|^2 = 1$ exactly is unaffected by this spatial jitter, the two observables are controlled by different mechanisms and $\mathcal{S}(0) < 0$. This does not violate any fundamental inequality: the bound $\mathcal{S} \geq 0$ established in Section V applies to a single ray, and the angular mixture of an extended source need not satisfy it.

The sensitivity parameter for $\delta g^{(2)}$ is $s = \sigma_0/\tau_c = \sqrt{2(1 - \Phi_{\rm eff})}\, \sigma_\phi(\Delta\lambda/\lambda)$; numerical values across α are tabulated in Table I.

F. Inversion Procedure for $r_\perp$

Within the small-s, narrow-band regime, the linear scaling $\delta g^{(2)} \approx 2\sqrt{2/\pi}\, \sigma_0/\tau_L$ together with the definition of σ_0 gives a direct relation between the measured zero-lag bunching suppression and the effective transverse coherence factor:

$$1 - \Phi_{\rm eff} = \frac{\pi}{16}\, \frac{(\delta g^{(2)})^2}{\sigma_\phi^2}\, \left(\frac{\lambda}{\Delta\lambda_L}\right)^2, \tag{51}$$

where $\Delta\lambda_L$ is the bandwidth of the Lorentzian component of the source line. The right-hand side contains only the directly observed quantities $\delta g^{(2)}$, $\Delta\lambda_L$, and λ, together with the foam variance $\sigma_\phi(D, \alpha)$ that is independently constrained by the point-source Siegert discrepancy $\mathcal{S}(\tau)$ at the same distance (Section V).

Once $\Phi_{\rm eff}$ is fixed by Eq. (51), Eq. (40) becomes a one-parameter equation for $r_\perp$ given the source brightness profile $I(\boldsymbol{\theta})$ (which is known independently from imaging or from the parent stellar model) and an assumed functional form for Φ. For a circularly symmetric source of angular diameter $\theta_{\rm src}$ and a kernel $\Phi(x)$ that decays on a scale of order

unity, $\Phi_{\rm eff}$ depends on the dimensionless ratio $\xi \equiv D\theta_{\rm src}/r_\perp$ alone, monotonically interpolating between $\Phi_{\rm eff} = 1$ at $\xi \ll 1$ (unresolved) and $\Phi_{\rm eff} \to 0$ at $\xi \gg 1$ (fully resolved); inverting numerically yields $r_\perp$. Limits of validity: Eq. (51) assumes the small-s expansion (Section VI E), the cross-ray Gaussian approximation (Section II D), and a spectrally resolved Lorentzian component $\Delta\lambda_L$; the breakdown at $D\theta_{\rm src} \sim r_\perp$ noted in Section II D caps the achievable precision in the intermediate angular-size regime.

Joint measurements of point-like and extended sources at the same redshift break the residual $(\sigma_\phi, r_\perp)$ degeneracy: the Siegert discrepancy probes σ_ϕ alone, while $\delta g^{(2)}$ probes the combination $\sigma_\phi^2\,(1 - \Phi_{\rm eff})$ via Eq. (51).

VII. DISCRIMINANTS AND SELF-CALIBRATION

The two principal observables developed in this paper — the point-source Siegert discrepancy $\mathcal{S}(\tau)$ of Section V and the extended-source zero-lag bunching suppression $\delta g^{(2)}$ of Section VI E — share two structural features that distinguish foam from conventional astrophysical phase screens, and a third feature that removes the need for absolute flux or source-spectrum calibration to detect a non-zero signal.

Distance scaling. The mathematical structure of the foam suppression factor $\mathcal{F}_{\rm foam}$ is identical to that of any Gaussian random phase screen, but the D-scaling is foam-specific: $\sigma_\phi^2 \propto D^{2(1-\alpha)}$ (D^1 for $\alpha = 1/2$, $D^{2/3}$ for $\alpha = 2/3$, D^0 for $\alpha = 1$), whereas a thin intervening screen scales as D and Kolmogorov turbulence as $D^{11/6}$; the holographic and conservative foam cases predict sub-linear or flat growth qualitatively different from any known astrophysical screen. $\mathcal{S}(\tau)$ inherits the full $D^{2(1-\alpha)}$ scaling through the variance factor σ_ϕ^2, while $\delta g^{(2)} \propto \sigma_\phi$ scales as $D^{1-\alpha}$; the resulting $2:1$ exponent ratio at fixed α provides the falsifiability test of Section II D.

Achromaticity. Geometrically, spacetime foam jitter ($\delta t = \Delta\ell/c$) is independent of λ, in contrast to plasma-based effects which scale as λ^2. As discussed in Section I, the existing LIV time-of-flight bounds do not constrain this achromatic jitter; the principal chromatic alternative the achromaticity discriminant rules out is therefore plasma propagation. The discriminant applies equally to both observables. We also note that for foam models with long-range transverse correlations, image blurring is strongly suppressed [15], so the coherence observables developed here may be the only channel sensitive to such models [3].

Self-calibration. Both observables are constructed as deviations from canonical values that hold for any Gaussian chaotic source: $\mathcal{S}(\tau)$ measures the departure of $g^{(2)} - 1$ from $|g^{(1)}|^2$ (zero in the absence of foam, regardless of the source spectrum), and $\delta g^{(2)}$ measures the departure of $g^{(2)}(0)$ from 2 (likewise zero in the absence of cross-ray jitter). *Detection* of a non-zero deviation therefore requires no separate flux calibration or source model. Quantitative interpretation does still depend on the source spectrum: the shape of $\mathcal{S}(\tau)$ retains a $|g^{(1)}_{\rm src}(\tau)|^2$ prefactor, and the magnitude of $\delta g^{(2)}$ depends on whether the line has a Lorentzian cusp at zero lag (Section VI E). Self-calibration mitigates the visibility–foam degeneracy at the detection stage; it does not eliminate source-model dependence in the inversion to foam parameters.

VIII. CONCLUSIONS

We have developed a theoretical framework for the imprint of spacetime foam — characterized by the Ng–Perlman exponent α — on both first- and second-order optical coherence. The central finding is that spacetime foam jitter produces two classes of observable violation of relations that hold for all Gaussian chaotic sources: a deviation from the Siegert relation, $g^{(2)}(\tau) - 1 = |g^{(1)}(\tau)|^2$, and a suppression of the zero-lag bunching peak below its canonical value of $g^{(2)}(0) = 2$. These second-order observables are complementary to the fringe visibility and Strehl ratio tests based on first-order coherence that have already been applied to constrain foam models. For the holographic model ($\alpha = 2/3$), all such observables lie well within the currently inaccessible parameter space: the first-order tests probe effects at the $\sim 10^{-8}$ level, while the second-order observables are characteristically smaller at $\sim 10^{-11}$ (a recent analysis of GRB 221009A claims consistency with holographic foam blurring [19], though without formal statistical significance). The predicted signals approach but do not clearly exceed the systematic floors of $\sim 10^{-9}$–10^{-10} anticipated for next-generation optical intensity interferometers, and detection would require both favorable foam parameters and significant advances in controlling detector systematics (Fig. 1).

The advantage of the second-order approach lies not in raw sensitivity but in two distinct features: the Siegert discrepancy is self-calibrating — any deviation of $\mathcal{S}(\tau) = g^{(2)}(\tau) - 1 - |g^{(1)}(\tau)|^2$ from zero is unambiguously non-Gaussian-chaotic, requiring no separate flux calibration or source model to detect; and for extended sources whose line profile has a Lorentzian component (natural or pressure broadening), the zero-lag bunching suppression $\delta g^{(2)}$ scales linearly with $s = \sigma_0/\tau_L$ — a consequence of the cusp in $g^{(2)}_{\rm src}$ at zero lag — so that despite being a second-order coherence measurement, $\delta g^{(2)}$ is not doubly suppressed relative to the first-order observables.

For the fiducial parameters $D = 1$ Gpc, $\lambda = 500$ nm, $\Delta\lambda/\lambda = 10^{-3}$, $A_\alpha = 1$: the $\alpha = 1/2$ case gives $\sigma_\phi \approx 280$ (fringes completely erased) and is already excluded by fringe-visibility measurements [12–14]. For the holographic model $\alpha = 2/3$, $\sigma_\phi \sim 10^{-8}$, so temporal point-source observables are negligibly small with current instrumentation. However, in the fully resolved limit $\Phi_{\rm eff} \to 0$ (corresponding to $D\Delta\theta \gg r_\perp$), the extended-source zero-lag bunching suppression scales linearly rather than quadratically with the small parameter $s = \sigma_0/\tau_c$ and yields $\delta g^{(2)} \approx 5.7\times10^{-11}$, evaluated in the dynamical limit $C_\alpha \to 0$ (a conditional upper bound). Since $s \propto \Delta\lambda/\lambda$, broader-band observations would increase $\delta g^{(2)}$ proportionally; for the holographic model the narrow-band validity condition $\sigma_\phi \cdot (\Delta\lambda/\lambda) \ll 1$ is trivially satisfied at any physically realizable bandwidth, so there is no approximation barrier to exploiting a broader bandpass, though a controlled calculation in the broadband regime — where the carrier/envelope decomposition underlying the Siegert violation is no longer rigorous — remains an open problem.

In any intensity interferometry measurement, a finite number M of collected spatiotemporal modes contributes a shot-noise floor $\delta g^{(2)}_{\rm floor} \sim M^{-1}$, and the foam detectability condition $\sigma_0/\tau_c \gtrsim M^{-1}$ should be quoted alongside any observational upper bound on $\sigma_\ell(D)$ derived from a non-detection.

A. Connection to Existing Radio-Astronomy Pipelines

The $g^{(2)}(0)$ observable developed here is already routinely computed in radio astronomy under different names: spectral kurtosis [24, 25], used for real-time RFI excision at the VLA, GBT, LOFAR, and CHIME, is constructed to estimate $g^{(2)}(0) - 1$ from channelized power samples (returning unity in expectation for a stationary Gaussian field, up to finite-sample bias corrections), as does the modulation index $m^2 \equiv \mathrm{Var}(I)/\langle I\rangle^2$, which equals $g^{(2)}(0) - 1$ for stationary, noise-corrected intensity samples and is used to characterize single-pulse intensity statistics of pulsars and FRBs [26]. The Siegert-violation framework developed here therefore translates directly to existing radio data products. The achievable sensitivity at radio wavelengths is set by the same combination $s \propto \sigma_\ell(D)\,\Delta\nu/c$ that controls the optical case, but with a modal shot-noise floor and systematic floor (RFI residuals, digitizer nonlinearity).

B. Cosmological Evolution and Future Directions

We have treated $\sigma_\ell^2(D)$ as static, justified by the negligible fractional change $H_0T \sim 10^{-10}$ over realistic observation durations. If α or A_α carries a redshift dependence, sources at different redshifts will not trace a simple power-law $\sigma_\phi(D)$, and the distance-scaling discriminants of Section VI generalize to a non-stationary, redshift-resolved framework that we leave to future work.

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